Selected kernel report

30 January 2026

Overview

This report summarizes selected kernels for the multi-config runs from `exp_022_mmr` through `exp_032_abess`. It includes kernel visualizations, kernel metadata, and the sign of each kernel's classifier weight (positive or negative).

Run configuration

Common settings: topM=300, response_fn=abs_max, batch size 64, lr=5e-4, epochs=60.

MMR runs

Run	Model	K	topM	std_feat	std_resp
exp_022_mmr	logistic	3	300	False	False
exp_023_mmr	logistic	5	300	False	False
exp_024_mmr	logistic	200	300	False	False
exp_025_mmr	logistic	300	300	False	False

ABESS runs

Run	Model	K	topM	std_feat	std_resp
exp_029_abess	logistic	3	300	False	False
exp_030_abess	logistic	5	300	False	False
exp_031_abess	logistic	200	300	False	False
exp_032_abess	logistic	300	300	False	False

Kernel sign summary

Run	K	Pos	Neg	Zero	Bias	Note
exp_022_mmr	3	3	0	0	-0.691	MMR
exp_023_mmr	5	5	0	0	-0.908	MMR
exp_024_mmr	200	126	74	0	-1.215	MMR
exp_025_mmr	300	181	119	0	-1.204	MMR
exp_029_abess	3	3	0	0	-0.649	ABESS
exp_030_abess	5	5	0	0	-1.004	ABESS
exp_031_abess	200	118	82	0	-1.267	ABESS
exp_032_abess	300	183	117	0	-1.224	ABESS

Notes A kernel is marked *positive* when its classifier weight is greater than zero, and *negative* when less than zero. Full kernel listings (rank, bank index, family, parameters, weight, sign) are saved to `Reports/figures/selected_kernels_summary.csv`.

Selected kernel visualizations

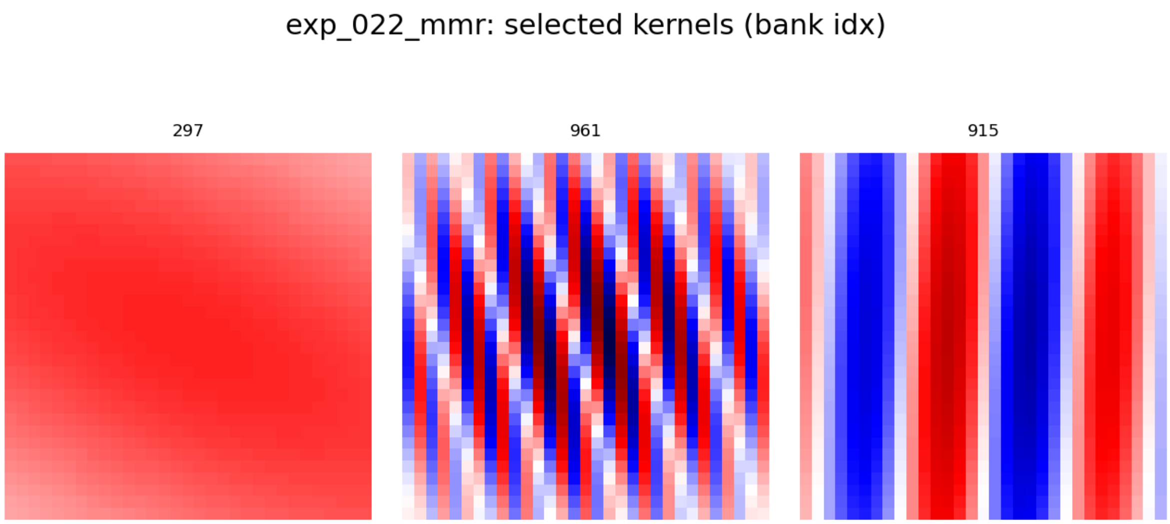

Figure 1: Selected kernels for exp_022_mmr (bank index shown above each kernel).

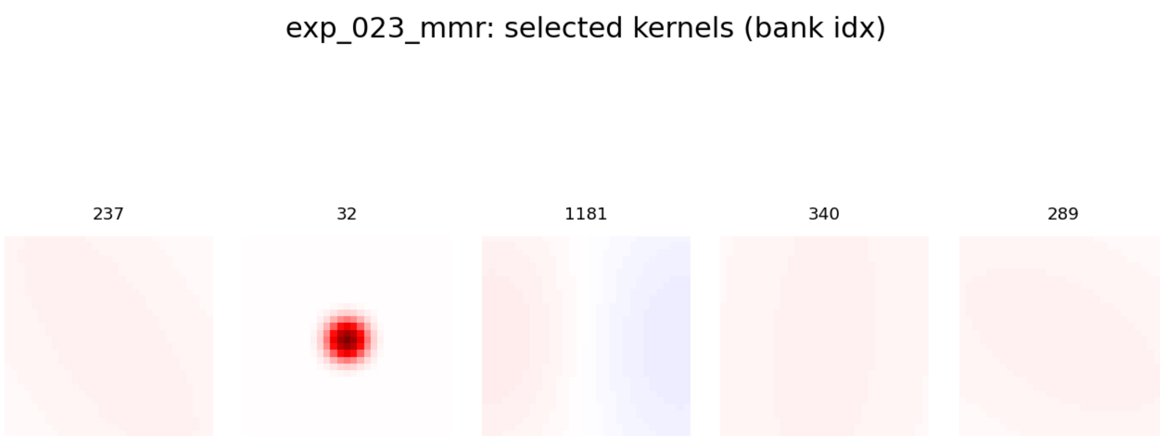

Figure 2: Selected kernels for exp_023_mmr (bank index shown above each kernel).

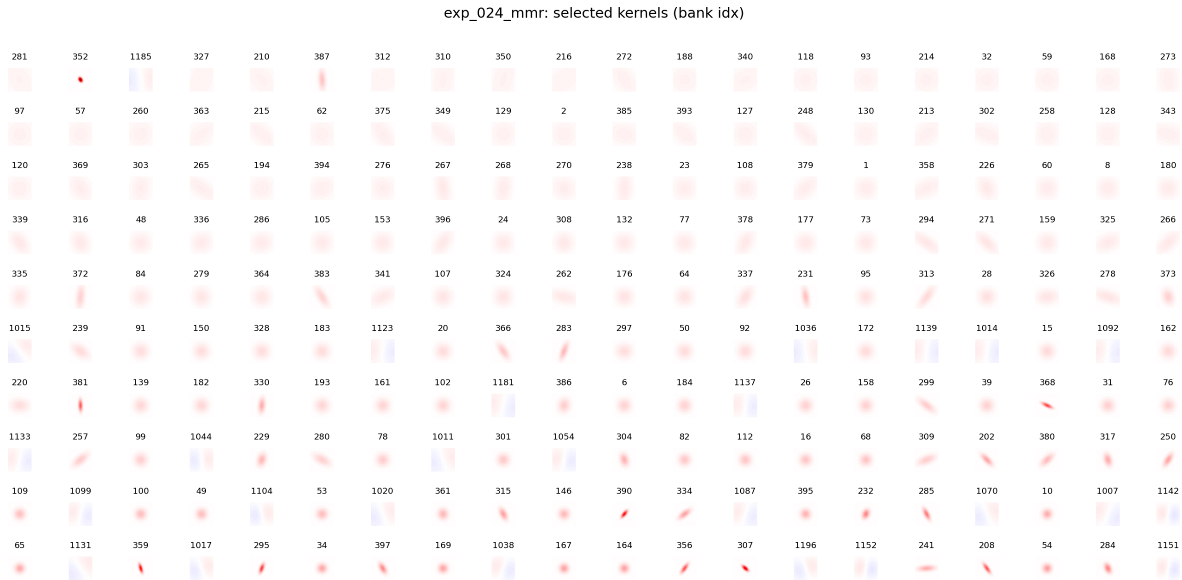

Figure 3: Selected kernels for exp_024_mmr (bank index shown above each kernel).

Figure 4: Selected kernels for exp_025_mmr (bank index shown above each kernel).

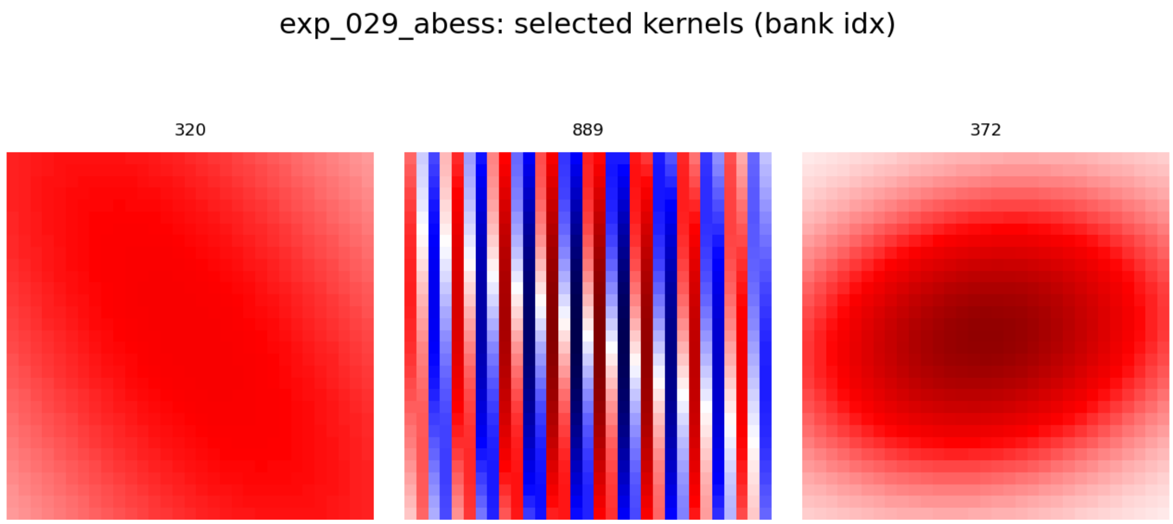

Figure 5: Selected kernels for exp_029_abess (bank index shown above each kernel).

Figure 6: Selected kernels for exp_030_abess (bank index shown above each kernel).

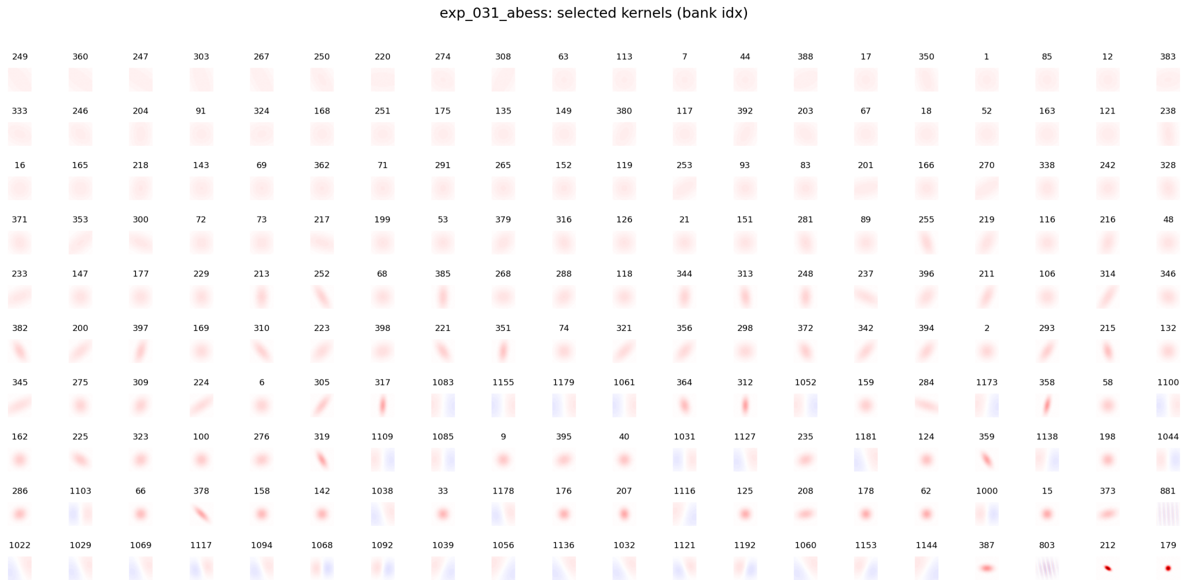

Figure 7: Selected kernels for exp_031_abess (bank index shown above each kernel).

Figure 8: Selected kernels for exp_032_abess (bank index shown above each kernel).

Top kernels by —weight—

The following tables list the top 10 kernels by absolute classifier weight for each run.

exp_022_mmr

Rank	Bank idx	Family	Params (abbrev)
2	961	gabor	sigma=12.303, freq=0.415, theta=0.054, phase=4.144, gamma=1.249, size=31
1	297	anisotropic_gaussian	sigma_x=14.856, sigma_y=38.192, theta=2.028, size=31
3	915	gabor	sigma=14.425, freq=0.072, theta=0.030, phase=1.380, gamma=0.737, size=31

exp_023_mmr

Rank	Bank idx	Family	Params (abbrev)	Sign	Weight
4	340	anisotropic_gaussian	sigma_x=15.104, sigma_y=42.629, theta=0.081, size=31	positive	0.8677
3	1181	hog	sigma=14.461, theta=0.023, size=31	positive	0.7312
5	289	anisotropic_gaussian	sigma_x=14.655, sigma_y=29.248, theta=2.112, size=31	positive	0.2675
1	237	anisotropic_gaussian	sigma_x=13.511, sigma_y=36.080, theta=2.552, size=31	positive	0.0258
2	32	gaussian	sigma_x=1.835, sigma_y=1.835, theta=3.115, size=31	positive	0.0094

exp_024_mmr

Rank	Bank idx	Family	Params (abbrev)	Sign	Weight
193	307	anisotropic_gaussian	sigma_x=1.514, sigma_y=3.128, theta=2.305, size=31	negative	-0.2338
2	352	anisotropic_gaussian	sigma_x=1.421, sigma_y=2.033, theta=2.610, size=31	negative	-0.2202
171	390	anisotropic_gaussian	sigma_x=1.564, sigma_y=3.657, theta=0.668, size=31	negative	-0.2013
175	232	anisotropic_gaussian	sigma_x=4.134, sigma_y=2.922, theta=1.976, size=31	negative	-0.1929
190	167	gaussian	sigma_x=4.029, sigma_y=4.029, theta=2.591, size=31	negative	-0.1695
183	359	anisotropic_gaussian	sigma_x=1.451, sigma_y=3.932, theta=2.861, size=31	negative	-0.1643
138	368	anisotropic_gaussian	sigma_x=1.589, sigma_y=4.734, theta=2.017, size=31	negative	-0.1616
174	395	anisotropic_gaussian	sigma_x=4.484, sigma_y=4.367, theta=0.759, size=31	negative	-0.1589
157	202	anisotropic_gaussian	sigma_x=2.558, sigma_y=6.329, theta=2.368, size=31	negative	-0.1555
199	284	anisotropic_gaussian	sigma_x=2.767, sigma_y=4.380, theta=2.711, size=31	negative	-0.1544

exp_025_mmr

Rank	Bank idx	Family	Params (abbrev)
289	174	gaussian	sigma_x=2.071, sigma_y=2.071, theta=1.774, size=31
3	283	anisotropic_gaussian	sigma_x=1.793, sigma_y=2.001, theta=1.804, size=31
290	395	anisotropic_gaussian	sigma_x=1.262, sigma_y=2.909, theta=0.560, size=31
251	367	anisotropic_gaussian	sigma_x=1.533, sigma_y=2.587, theta=2.873, size=31
299	275	anisotropic_gaussian	sigma_x=1.646, sigma_y=2.978, theta=1.300, size=31
49	858	gabor	sigma=11.845, freq=0.130, theta=3.133, phase=1.695, gamma=1.221, size=31
270	63	gaussian	sigma_x=2.302, sigma_y=2.302, theta=0.141, size=31
280	119	gaussian	sigma_x=2.626, sigma_y=2.626, theta=1.881, size=31
11	318	anisotropic_gaussian	sigma_x=14.402, sigma_y=23.362, theta=0.320, size=31
10	203	anisotropic_gaussian	sigma_x=14.816, sigma_y=39.076, theta=1.204, size=31

exp_029_abess

Rank	Bank idx	Family	Params (abbrev)
3	372	anisotropic_gaussian	sigma_x=11.561, sigma_y=7.902, theta=2.940, size=31
1	320	anisotropic_gaussian	sigma_x=14.951, sigma_y=37.142, theta=2.537, size=31
2	889	gabor	sigma=12.113, freq=0.255, theta=3.112, phase=1.578, gamma=0.736, size=31

exp_030_abess

Rank	Bank idx	Family	Params (abbrev)
2	350	anisotropic_gaussian	sigma_x=15.312, sigma_y=44.003, theta=2.073, size=31
5	916	gabor	sigma=14.013, freq=0.263, theta=3.090, phase=4.547, gamma=1.460, size=31
1	259	anisotropic_gaussian	sigma_x=15.129, sigma_y=40.973, theta=2.769, size=31
3	246	anisotropic_gaussian	sigma_x=13.684, sigma_y=34.884, theta=2.458, size=31
4	294	anisotropic_gaussian	sigma_x=14.403, sigma_y=31.963, theta=0.795, size=31

exp_031_abess

Rank	Bank idx	Family	Params (abbrev)	Sign	Weight
199	212	anisotropic_gaussian	sigma_x=2.611, sigma_y=1.372, theta=0.543, size=31	negative	-0.2458
200	179	gaussian	sigma_x=1.929, sigma_y=1.929, theta=0.542, size=31	negative	-0.2216
197	387	anisotropic_gaussian	sigma_x=3.175, sigma_y=5.504, theta=1.605, size=31	negative	-0.1909
157	359	anisotropic_gaussian	sigma_x=3.129, sigma_y=6.639, theta=2.594, size=31	negative	-0.1453
175	178	gaussian	sigma_x=4.833, sigma_y=4.833, theta=0.030, size=31	negative	-0.1447
3	247	anisotropic_gaussian	sigma_x=13.770, sigma_y=40.930, theta=2.170, size=31	positive	0.1352
24	91	gaussian	sigma_x=13.807, sigma_y=13.807, theta=1.855, size=31	positive	0.1350
7	220	anisotropic_gaussian	sigma_x=14.050, sigma_y=33.972, theta=1.584, size=31	positive	0.1334
13	44	gaussian	sigma_x=15.181, sigma_y=15.181, theta=0.638, size=31	positive	0.1333
32	117	gaussian	sigma_x=13.372, sigma_y=13.372, theta=3.007, size=31	positive	0.1326

exp_032_abess

Rank	Bank idx	Family	Params (abbrev)	Sign	Weight
288	119	gaussian	sigma_x=2.039, sigma_y=2.039, theta=2.935, size=31	negative	-0.2167
287	162	gaussian	sigma_x=2.098, sigma_y=2.098, theta=1.138, size=31	negative	-0.1893
257	398	anisotropic_gaussian	sigma_x=1.240, sigma_y=2.948, theta=2.801, size=31	negative	-0.1889
284	196	gaussian	sigma_x=2.142, sigma_y=2.142, theta=0.274, size=31	negative	-0.1872
251	355	anisotropic_gaussian	sigma_x=2.658, sigma_y=1.767, theta=0.950, size=31	negative	-0.1871
274	51	gaussian	sigma_x=2.674, sigma_y=2.674, theta=0.687, size=31	negative	-0.1797
279	153	gaussian	sigma_x=2.634, sigma_y=2.634, theta=2.281, size=31	negative	-0.1794
300	321	anisotropic_gaussian	sigma_x=2.653, sigma_y=1.500, theta=2.616, size=31	negative	-0.1768
246	399	anisotropic_gaussian	sigma_x=1.445, sigma_y=3.069, theta=3.035, size=31	negative	-0.1745
285	94	gaussian	sigma_x=2.172, sigma_y=2.172, theta=2.690, size=31	negative	-0.1724